STOCKPORT COUNTY FC
RECRUITMENT MODEL SPECIFICATION

Decision-Grade Recruitment Intelligence System

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Revised to address: process control, scoring mechanics, position-specific age logic, entity-level source hierarchy, outcome framework, governance strictness, implementation realism

Section 1 — Model Purpose

This model exists to make Stockport County's recruitment decisions sharper, faster, and more accountable. It does not replace scouts, coaches, or the sporting director. It improves the information those people have when they make decisions.

Five decisions this model serves:

1. **Who should we scout next?** Narrows thousands of candidates to 30–50 evidence-backed names per brief, filtered for affordability, age profile, and Championship projection.
2. **Does this target fit our game model?** Scores role-fit against Stockport's tactical system. A centre-back who cannot play out from the back scores poorly regardless of defensive stats.
3. **Can this player survive promotion?** Projects League One output into Championship-equivalent performance. Flags players whose numbers only work at this level.
4. **Is this player worth the fee?** Compares cost against expected output, availability risk, resale timeline, wage fit, and comparable transactions.
5. **Should we sign externally or develop from within?** Compares external targets against internal pathway players on the same metrics. Flags when an academy player is approaching readiness.

1.1 The Three Tensions Stockport Faces

Stockport is not a Premier League club. It is not a survival-focused Championship side. It occupies a specific strategic position: ambitious, ascending, financially disciplined, asset-conscious, building toward the Championship as a sustained destination. That demands a model designed around three tensions:

- **Immediate output vs. future value.** Every signing must help now, but the club cannot fill the squad with depreciating assets. The model scores both and presents them transparently.
- **League One dominance vs. Championship readiness.** A player who dominates League One but cannot scale is a trap. A player who projects to Championship level but cannot help win promotion is premature.
- **Spending to compete vs. spending to build.** Recruitment must contribute to asset growth, not just squad strength. A £200k fee for a 22-year-old who could be worth £1m is different from the same fee for a 29-year-old.

Section 2 — Recruitment Decision Framework

Every signing passes through eight stages. The model has a defined role at each stage. It never owns the decision.

Stage	Key Inputs	Model Output	Decision Owner	Success
1. Brief Creation	Positional need, game model, budget, timeline	None — brief is human-authored	HoR + Coach	Brief is specific, approved, and resourced
2. Market Scan	Live brief + league scope	Universe of candidates passing hard gates	Recruitment Analyst	Wide net; no good candidate missed
3. Longlist	Scan results + role-fit scoring	Ranked 40–60 players with scores + projections	Analyst	Every name has a data-backed reason
4. Fit + Value Filter	Longlist + coach input + finance parameters	Filtered list with tactical flags + value ranking	HoR + Coach + Finance	Coach confident in system fit; finance confident in viability
5. Shortlist	Filtered list + availability risk + pathway comparison	8–12 prioritised names	HoR	Shortlist is defensible and actionable
6. Video + Live Scouting	Shortlisted players	Key matches, statistical context, zone data	Scouts + Analyst	Scouts watch right matches with right questions
7. Cross-Functional Review	Scout reports + data pack + finance + medical	Transfer readiness pack	HoR + Coach + Finance + Medical	Every stakeholder concern addressed
8. Final Recommendation	Cross-functional consensus	Board recommendation brief	Sporting Director / CEO	Board approves with confidence; audit trail exists

2.1 Process Control Requirements

This is the section that makes the framework operational, not just descriptive.

2.1.1 Required Inputs Before a Brief Becomes Live

No market scan begins until all of the following are documented and signed off by HoR:

- **Role definition:** Specific role (not position), approved by first-team coaching staff. ‘We need a centre-back’ is not a brief. ‘We need a left-sided ball-playing CB who can progress play under pressure in a 4-3-3 build-up’ is a brief.

- **Budget envelope:** Maximum fee, maximum annual wage, maximum contract length. Approved by finance.
- **Archetype and intent:** Primary archetype (Section 3) and recruitment intent. Is this a promotion push, a Championship-building signing, a development investment, or short-term cover?
- **Age parameters:** Acceptable age range, justified by archetype and role.
- **League scope:** Which leagues to search. Default: all tracked. Restricted if budget or registration limits apply.
- **Timeline:** Target completion date. A January window target has different urgency from a summer planning target.
- **Pathway check:** Confirmation from academy manager whether any internal player is within 18 months of readiness at this role. If yes, the pathway comparison is mandatory from the outset.

Non-Negotiable

If any of these inputs is missing, the brief is not live and no analyst time is spent on it. This prevents speculative searches that consume resource without clear intent.

2.1.2 Minimum Scouting Requirement Before Cross-Functional Review

No player progresses to Stage 7 (Cross-Functional Review) unless:

- **Minimum 3 matches reviewed on video** by a scout or analyst, with structured notes per match (technical/tactical/physical/mental ratings 1–5 plus free text).
- **At least 1 match reviewed must be against top-6 opposition** in the player's current league (tests performance under pressure, not just against weaker sides).
- **At least 1 live observation OR video review by the first-team coach** confirming system-fit assessment. Coach must provide written sign-off: 'fits our system' / 'concerns about [specific aspect]' / 'does not fit'.
- **Wyscout zone-filtered analysis completed** for any player being evaluated as Championship Transition or Emerging Asset archetype (deep scouting enrichment from Tier 2 layer).

Exception: For urgent January window targets where time is compressed, HoR can waive the live observation requirement but must document this as a risk factor in the recommendation pack.

2.1.3 Conditions for Longlist to Shortlist Transition

A player moves from longlist (Stage 3) to shortlist (Stage 5) only when ALL of the following are met:

6. Passes all applicable hard gates (Section 4)
7. Role-fit score ≥60th percentile against the role template (or analyst override with documented justification)
8. Confidence tier is Medium or above (≥10 tracked appearances)
9. For Championship Transition and Emerging Asset archetypes: Championship projection base case (50th percentile) is at or above 40th percentile for that role at Championship level
10. Financial value assessment completed: fee estimate, wage fit, and resale band documented
11. Internal pathway comparison completed: if a pathway player exists at this role, the external target must score meaningfully higher (≥15% composite advantage) to justify the spend

If any condition is not met: the player remains on the longlist for monitoring. They can be promoted later if conditions change (e.g., more data becomes available, price drops, pathway player is injured).

2.1.4 Definition of a Failed Recommendation

A recommendation is classified as failed if, 18 months post-signing, ANY of the following apply:

Failure Type	Definition	Severity
Performance failure	Player's per-90 output in role-defining metrics is below 30th percentile at the destination level for 2+ consecutive assessment windows (6-month periods)	High
Availability failure	Player available for <60% of matchdays in the first 18 months (excluding freak injuries documented as unforeseeable)	High
Financial failure	Player's market value has declined by >40% from acquisition cost within 18 months AND performance does not justify retention	High
System failure	Player has been unable to integrate into the game model and has been moved to a different role or excluded from the preferred XI despite being fit	Medium
Pathway cost	Signing directly blocked a named pathway player who subsequently left the club or stalled in development as a documented consequence	Medium
Overpayment	Fee paid was >30% above the financial model's fair-value band ceiling, and the player has not outperformed the base-case projection	Medium

What failure triggers: A post-mortem review. Was the failure predictable from the data? Was it flagged as a risk and overridden? Was the data wrong? Was the model wrong? Was the scouting wrong? The answer feeds back into model improvement, weight calibration, and process tightening. Failure is not blame. It is learning.

Critical Distinction

A player who was signed with acknowledged risk (e.g., High availability risk, accepted by medical) and then gets injured is NOT a model failure. The risk was identified and accepted. A player who was presented as low-risk and then gets injured IS a model failure if the risk was knowable from the data. The failure classification must distinguish between accepted risk and missed risk.

Section 3 — Player Archetype Strategy

Archetypes are not mutually exclusive buckets. Real players often straddle categories. The system assigns a primary archetype, an optional secondary archetype, and a recruitment intent.

3.1 Archetype Definitions

Dimension	Promotion Accelerator	Championship Transition	Emerging Asset	Low-Cost Bridge
Purpose	Win promotion THIS season	Perform at Championship level	Develop into valuable player or saleable asset	Reliable depth at minimal cost
Typical age	24–29	22–27	18–22	Any (often 28+)
Cost logic	Fair market; accept limited resale	Moderate fee; value growth expected	Low/free; upside is development	Free/loan/minimal
Risk tolerance	Low: must deliver immediately	Medium: adaptation risk accepted	Higher: 40–50% may not reach level	Very low: must not disrupt
Resale expectation	Low	High	Highest	None
Time horizon	1–2 seasons	2–4 seasons	3–5 seasons	6–18 months

3.2 Dual-Archetype Assignment

Every target is assigned:

- **Primary archetype:** The main reason for signing. Determines evaluation weights.
- **Secondary archetype (optional):** A legitimate second purpose. Affects how risk is assessed.
- **Recruitment intent:** A one-sentence statement of why this specific player for this specific role at this specific time.

Examples of valid dual assignments:

- A 24-year-old midfielder: Primary = Promotion Accelerator, Secondary = Championship Transition. Intent: ‘Start 35+ matches to drive promotion; project to remain a starter at Championship level.’
- A 21-year-old winger: Primary = Emerging Asset, Secondary = Promotion Accelerator. Intent: ‘Immediate squad rotation with 15–20 starts; expected to develop into first-choice winger within 18 months.’
- A 27-year-old centre-back: Primary = Promotion Accelerator, Secondary = Championship Transition. Intent: ‘Leadership spine for promotion push; still holds value if club moves up due to Championship-level experience.’

Why this matters: The dual assignment determines which scoring weights apply (Section 5). A player evaluated purely as a Promotion Accelerator gets heavy weighting on current performance. But if the secondary archetype is Championship Transition, the upward projection score also contributes. This prevents the model from ignoring long-term value when evaluating someone who also needs to deliver now.

Archetype Distribution Target

Approximate squad balance: 5–7 promotion accelerators (spine), 6–8 Championship transition (future core), 3–5 emerging assets (pipeline), 4–6 bridge signings (depth). If the squad tips too far toward bridge signings, the club is surviving, not building. If too far toward emerging assets, it risks lacking experience to win promotion.

Section 4 — Hard Gates

Gates are split into three classes. This prevents over-rigid filtering that narrows the market too early while still enforcing non-negotiable standards.

Gate	Class	Rule	Override / Exception
Registration eligibility	Absolute Exclusion	Must be registrable under EFL rules. International visa/permit must be obtainable.	None. Cannot be overridden.
Budget affordability	Absolute Exclusion	Total cost (fee + wages × contract) within approved brief budget.	CEO/board can extend budget — new brief required.
Role relevance	Absolute Exclusion	Must map to an active, approved brief.	None. No speculative additions.
Pathway blocking (hard)	Absolute Exclusion	If internal player is <12 months from readiness at same role AND external target is same age or younger on 3+ year deal.	Academy Manager + HoR joint sign-off only.
Wage discipline	Strong Caution	Proposed salary should not exceed 120% of band for role. Flagged if exceeded.	HoR + Finance approval. Documented as elevated financial risk.
Age-profile fit	Strong Caution	Should fit archetype age range. Flagged if outside range.	HoR approval with role-specific justification. CBs and GKs have wider acceptable ranges than wingers.
Availability risk (High)	Strong Caution	≥40% chance of missing >25% of matches. Flagged prominently.	Medical department clinical assessment required before progression. If medical clears, player proceeds with risk documented.
Data confidence (Low)	Strong Caution	<10 tracked appearances. Insufficient data for reliable scoring.	Scout recommendation + HoR approval to add to monitoring track (not shortlist).
Championship projection (below threshold)	Exception Requiring Sign-Off	For Championship Transition / Emerging Asset archetypes: base-case projection below 40th percentile.	HoR + Coach joint sign-off if role is depth-only or specific system need. Documented as projection risk.
Pathway blocking (soft)	Exception Requiring Sign-Off	Internal player exists at role but is 12–24 months from readiness. External target on 2+ year deal.	Discussed at recruitment meeting. No hard gate but pathway impact assessment required.

Application order: Absolute Exclusions first (instant disqualification, zero analyst time). Then Strong Caution flags attached to surviving candidates. Then Exception checks during shortlisting phase. This minimises wasted effort while preserving flexibility.

Role-sensitivity note: Age gates must account for positional age curves. A 29-year-old centre-back is mid-peak. A 29-year-old winger is entering decline. A 29-year-old goalkeeper may be 3 years from peak. The Strong Caution flag on age-profile is evaluated against the position-specific curve from Section 6, not a universal threshold.

Section 5 — Scoring Framework

5.1 The Six Layers

Layer	What It Measures	Why at Stockport	Measure	Do NOT Over-Weight
1. Tactical Fit	System compatibility	Wrong-system players are expensive mistakes	Pressing profile, build-up involvement, defensive shape	Raw output without system context
2. Role Fit	Statistical match to specific role template	Positions are too broad; roles are what matter	Weighted similarity to role template	Position label alone
3. Current Performance	Output quality at current level right now	Must help the team compete this season	League-adjusted per-90 percentiles, consistency, form	Season totals; stats vs. weak opposition
4. Upward Projection	Can they perform at Championship?	Separates building from surviving	Translation model output, trend, age curve, vs. top-6	Current output alone as proxy for future
5. Financial / Asset	Is this financially rational?	Asset growth is core to strategy	Fee vs. comps, wage fit, resale band, replacement cost	Market value alone
6. Availability & Risk	Will they actually be available?	Unavailable players cost more per appearance	3yr rate, injury profile, recurrence, fitness run	Single-season data

5.2 Layer Interaction Mechanics

This section defines how layers combine. Two different analysts building from this document should produce materially similar models.

Mechanic	Rule	Rationale
Tactical Fit acts as a cap	If Tactical Fit < 40th percentile, composite score is capped at 60 regardless of other layers. The player does not fit the system.	Prevents statistically strong players in wrong systems from ranking highly.
Availability Risk is subtractive	Availability risk score is converted to an availability discount: (availability_rate / 100). Applied multiplicatively to the composite. A player available 70% of the time gets 0.70 × composite.	Directly models the economic reality: a player who misses 30% of matches costs 30% more per appearance.
Financial / Asset Value is multiplicative via VAR	Composite quality score is divided by normalised cost to produce a Value-Adjusted Ranking. VAR =	Allows ranking by value-for-money alongside pure quality.

	quality_composite / (total_cost / league_avg_cost_for_role). Above 1.0 = good value.	
Upward Projection can override Current Performance for specific archetypes	For Emerging Asset archetype: if projection (50th percentile base case) is ≥70th percentile at Championship level, current performance floor can be lowered to 30th percentile (from default 50th). Player is bought for the future.	Prevents the system from rejecting young players whose current output is developing but whose trajectory is strong.
Archetype weighting is fixed per window phase	Weights are set at brief creation and do not change during the window. Mid-window weight changes would re-rank the shortlist and cause confusion.	Ensures consistency within a recruitment cycle.
Dual-archetype scoring	Primary archetype weights contribute 70%. Secondary archetype weights contribute 30%. Blended composite reflects the dual purpose.	A player who is both a promotion accelerator and a Championship transition player is scored as 70% immediate + 30% future.

5.3 Archetype Weighting Table

Layer	Promotion Accelerator	Championship Transition	Emerging Asset	Low-Cost Bridge
Tactical Fit (cap logic applies)	25%	20%	15%	15%
Role Fit	20%	20%	15%	20%
Current Performance	25%	15%	10%	25%
Upward Projection	5%	25%	30%	5%
Financial / Asset (via VAR)	10%	15%	25%	20%
Availability (multiplicative discount)	15%	5%	5%	15%

5.4 Score Presentation Rules

- **Never present a single composite score in isolation.** Always show per-layer breakdown.
- **Always show the confidence tier.** Low-confidence scores are visually distinct (greyed out, asterisked).
- **Always show projections as ranges (10th–50th–90th percentile).** Never as point estimates.
- **Always show the internal pathway comparison** if an internal player exists at the same role.
- **Always show which archetype weights were applied** and whether dual-archetype blending was used.
- **Always flag the tactical fit cap** if it constrained the composite score.

Section 6 — Age-Value Strategy

6.1 Position-Specific Age Curves

Age impacts different positions differently. The model must evaluate age against role-specific curves, not universal bands.

Position Group	Physical Peak	Output Peak	Decline Onset	Resale Window	Acceptable 'Older' Signing Age
Goalkeeper	27–32	28–34	34–36	25–30	Up to 33 (long peak window)
Centre-back	26–31	27–32	32–34	24–29	Up to 31 (reliability still high)
Full-back / Wing-back	24–29	25–30	30–32	22–27	Up to 29 (physical demands high)
Central midfielder	25–30	26–31	31–33	23–28	Up to 30 (engine longevity varies)
Wide attacker / Winger	23–28	24–29	29–31	21–26	Up to 28 (pace-dependent decline earlier)
Striker	24–29	25–30	30–32	22–27	Up to 29 (movement can compensate for pace loss)

6.2 Role-Specific Resale Logic

Contract length interacts with age to determine resale viability:

Scenario	Resale Probability	Recommended Contract
22-year-old CB on 4-year deal	Very High. 3 years to build value, 1 year to sell with leverage.	4 years. Maximum value window.
25-year-old winger on 3-year deal	Medium. At or near peak. 1–2 years of peak output, then declining value.	3 years maximum. Sell at 26–27 if not promoting.
28-year-old striker on 3-year deal	Low. Depreciating from day one. Contract will be a liability by year 3.	2 years (1+1 option preferred). No long-term commitment.
21-year-old GK on 5-year deal	High but slow. GK development is long. May not see returns for 3–4 years.	4–5 years. Patience required. Long peak window justifies.
27-year-old CM on 3-year deal	Medium-Low. Near peak but CM resale depends on profile. Versatile/athletic CMs hold value longer.	3 years. Acceptable if promotion push; short if bridge signing.

6.3 The 22–24 Sweet Spot

This is Stockport's competitive advantage band. Old enough to contribute now. Young enough to grow in value. Bigger clubs overlook this band for established names. Smaller clubs cannot afford it. Stockport should concentrate spending here for Championship Transition signings.

6.4 Avoiding the Youth Trap

- **Genuine upside:** Young AND showing upward trajectory AND in competitive environment. Real emerging asset.
- **Raw youth without trajectory:** Young but same output for 2+ seasons with no improvement. Being 20 does not make a player valuable.
- **Hype without sample:** Few eye-catching performances but insufficient data. Confidence gates must apply. Age alone does not override data uncertainty.

Section 7 — League Transition Projection

The most dangerous mistake in lower-league recruitment: signing players who look dominant at the current level but do not scale. This section defines how to avoid it.

7.1 Three-Dimensional Projection

Projection is not one number. The model outputs three separate estimates:

Dimension	What It Predicts	Model Approach	Why It Matters Separately
Projected Performance	Expected per-90 output at Championship level for role-defining metrics	Hierarchical regression with league-pair effects, controlling for age, role, team strength, usage	A player may be capable of good per-90s but only in limited minutes
Projected Minutes Share	Expected % of available matchdays the player would start at Championship level	Logistic regression on historical movers: did they become starters (≥25 starts) in first Championship season?	A player who projects as a 15th-percentile starter is a squad player at best, regardless of per-90 quality
Projected Adaptation Time	Expected months before reaching steady-state output at new level	Historical first-season vs. second-season output for L1 → Championship movers. Estimate the dip and recovery curve.	A player who needs 12 months to adapt may miss the promotion window entirely

7.2 Adjustment Factors

Factor	What It Captures	Typical L1 → Championship Impact
League strength translation	Raw output declines at stronger level	-10% to -25% on most per-90 metrics
Team-dominance effect	Dominant L1 team inflates stats; Championship opponents won't be dominated	-5% to -15% for players at top-3 L1 teams
Usage inflation	90-minute starter at L1 may share time at Championship	Per-90 may hold but total contribution drops
Physical and tempo	Championship is faster; players must sustain higher intensity	Affects low-physical profiles most
Role scalability	Some roles function differently at higher levels	Target men may get less service; controllers face more pressure
Adaptation dip	First-season performance typically below steady-state	Expect 5–10% below projected steady-state in year 1

The Most Dangerous Player Profile

A 26-year-old forward at a dominant League One team averaging 0.7 goals per 90 with an xG of 0.4. Massively overperforming xG, playing for a team creating chances 40% above league average, against structurally weaker defences. Projected Championship output after adjusting for league strength, team dominance, xG regression, and adaptation dip: 0.20–0.35 goals per 90. Well below Championship average. The model must catch this player before the club offers £500k.

7.3 Honest Limitations

This section must be transparent about what the projection can and cannot do:

- **Training data is thin for lower-league transitions.** L1 → Championship has reasonable volume. L2 → Championship is sparse. The model must report the sample size per league pair and flag low-sample projections.
- **Survivorship bias is real.** Only players who were bought (i.e., deemed good enough) appear in the training data. The model learns from a positively selected population. Projections for borderline players may be optimistic.
- **Decision-making speed cannot be modelled from data.** This is a scouting-only assessment. The model flags it as a required video/live evaluation, not a data output.
- **Projections beyond 2 levels are unreliable.** L2 → Championship is a two-level jump. The model should express very wide uncertainty bands and recommend caution.

Section 8 — Role-Based Modelling

Role Family	Roles	Key Metrics	Stockport Context
Centre-back	Ball-playing CB, Aggressive CB, Covering CB	Pass accuracy, progressive passes, tackles vs. interceptions, aerial rate, dribbled past	System determines which CB profile matters. Ball-playing CBs if playing out from back.
Full-back / Wing-back	Attacking FB, Inverted FB, Defensive FB, Wing-back	Key passes, crosses, dribbles, progressive carries vs. tackles, interceptions	Critical role distinction. Wing-back in 3-5-2 is entirely different from defensive FB in 4-4-2.
Midfield Controller	Destroyer, Deep playmaker, Ball-winner	Tackles, interceptions, pass volume/accuracy, progressive passes	Defines build-up identity. Template must match coach preference.
Transition Midfield	Box-to-box, Mezzala, Ball-carrying 8	Balanced: tackles + shots + dribbles + duels + distance	Engine of promotion-pushing side. Must score on BOTH attack and defence.
Wide Creator / Runner	Inside forward, Traditional winger, Wide playmaker	Shots + dribbles + key passes (IF) vs. crosses + carries (winger)	Must not compare inside forwards against crossing wingers on same template.
Striker	Target, Pressing forward, Poacher, Complete, False 9	Aerial rate, fouls drawn, pressing actions, shots/90, xG/shot, goals vs. xG	Attacking system determines which profile fits.
Pressing overlay	Applies across all outfield roles	Forward defensive actions/90, recoveries, pressing intensity	Minimum pressing threshold for all outfield players in a pressing system.

8.1 Role Template Governance

- **Core templates are stable:** 8–12 metrics per role with importance weights. Reviewed quarterly by coaching staff.
- **Sub-profiles are variable:** Within a core template, weights can be adjusted for specific opponent preparations or game-model phases. These are logged as temporary variants, not permanent changes.
- **Formal approval log:** Every template change is versioned with date, author, and justification. The analyst building the model can always trace which template version produced which scores.
- **Clustering is an aid, not the truth:** If k-means clustering is used to infer roles, the clusters must be validated against coaching staff's tactical definitions. Tidy clusters that do not align with football reality are discarded.

Section 9 — Internal Pathway Integration

External recruitment and internal development are connected, not separate. Every shortlist automatically includes pathway players at the relevant role.

9.1 Pathway Comparison as a Decision Prompt

The pathway comparison is a prompt for conversation, not a mathematical substitute for development judgement.

When an internal player appears alongside external targets, the system presents:

- The internal player's current per-90 profile scored against the same role template
- Their trend slope over the last 6–12 months (improving, stable, or declining)
- The gap between their current output and the external target's output
- An estimated readiness window: how long until they could plausibly reach the external target's current level, based on trajectory

What the system does NOT do: It does not assign a binary 'ready' / 'not ready' label based on a fixed threshold. Readiness is not just output. It is also physical maturity, tactical reliability, personality, training consistency, ability to absorb setbacks, and staff trust. These are human-assessed and documented in the academy manager's development report, not inferred from data.

> **Decision Rule**
> If the data shows an internal player is within 15% of the external target on role-fit and projection, the recruitment meeting MUST include the academy manager and explicitly discuss: (a) can we wait? (b) what is the cost of waiting? (c) what is the cost of signing over this player? That discussion may still result in the external signing. But the conversation must happen.

9.2 Loan Strategy Support

- Track loaned players on the same per-90 metrics and role templates as external targets
- Flag when loaned player's output suggests they are outgrowing their loan level
- Track loan destination quality: minutes played, level of competition, role usage
- Loan recall is a development decision, not a data decision. The system surfaces information; the academy and coaching staff decide.

Section 10 — Data Architecture

Each data entity has its own source hierarchy. There is no blanket ‘canonical source’. Different entities require different sources of truth.

Entity	Source of Truth	Fallback	Confidence	Reconciliation Rule	Must NOT Be Used For
Per-match player stats	API-Football /fixtures/players	Wyscout Advanced Search	High	If sources disagree by >10% on any per-90: investigate. Use API-Football unless proven incorrect.	Do not use season aggregates when match-level data exists
Match events / timeline	API-Football /fixtures/events	Wyscout match XML	High	API-Football is canonical. Wyscout supplements for zone context.	Do not use event data to infer physical metrics
Team match context	API-Football /fixtures/statistics	None	High	Single source. If missing for a fixture, flag fixture as partial coverage.	Do not impute missing possession data
Lineups / formations	API-Football /fixtures/lineups	Wyscout	High	API-Football canonical. Cross-check formation with Wyscout if discrepancy.	Do not use grid positions as precise location data — they are approximate
Player bio (age, height, nationality)	API-Football /players	Transfermarkt	High	If DOB disagrees: use API-Football. If height disagrees by >3cm: flag for manual check.	Do not use bio data from news articles or agent profiles
xG, xA, progressive actions	FBref (StatsBomb)	Proxy xG model	Med-High (FBref); Medium (proxy)	FBref is truth where available. Proxy model for uncovered leagues. Never mix sources for same player.	Do not present proxy xG with same confidence as FBref xG
Aerial duels (separated)	Wyscout or FBref	Proxy from height + duels	Med-High (Wyscout); Low (proxy)	Wyscout is preferred. Proxy flagged prominently.	Do not use proxy aerial data in composite scoring without discount
Transfer history	API-Football /transfers	Transfermarkt	High	API-Football for dates/clubs. Transfermarkt for fees (API-Football lacks fees).	Do not treat Transfermarkt fees as exact — they are estimates
Injury / sidelined	API-Football /sidelined + /injuries	Club medical (own players)	Medium (external); High (internal)	External data may miss minor injuries. Internal records are truth for own players.	Do not assume absence of injury record means player is injury-free
Market value	Transfermarkt	None	Medium (lag, crowd-sourced)	Single source. Treat as indicative, not precise. Always	Do not use market value as a fee prediction — actual fees vary significantly

				present as approximate.	
Contract / wages	Transfermarkt (partial)	Club records (own players)	Low (external); High (internal)	External contract data is often incomplete. Flag as estimated.	Do not make financial decisions based on estimated external wages alone
Zone-segmented data	Wyscout zone-filtered export	None	Med-High (pipeline players only)	Manual export per player. Timestamp and flag if >30 days old.	Do not extrapolate zone data to players not individually analysed
Physical tracking	League tracking provider	Wyscout physical layer / club GPS	High (where available)	Preferred: broadcast tracking. Club GPS for own players. Flag source per player.	Do not compare GPS data across different providers without calibration
Character / mentality	Scout reports + player care only	None	Scouting-only	NEVER scored. NEVER in automated systems.	Do not create data proxies for character from match data
Pathway player data	Club academy performance system	None	Highest (own players)	Internal source is truth. External benchmarking via API-Football for context.	Do not compare internal player data quality to external — different granularity

10.1 Handling Missing Data and Low-Sample Players

- **Missing data:** Store as null. Never impute as zero. Flag in all outputs. Apply confidence discount.
- **Low-sample players (<10 appearances):** Shrink per-90 estimates toward position-league mean. Classify as Low confidence. Exclude from automated shortlisting. Can be added to monitoring track via scout recommendation.
- **Coverage gaps:** Some L1/L2 clubs have poor event data. Track coverage quality per league. Flag leagues below 80% stat completeness.

Section 11 — Model Stack

Seven components. Each has explainability requirements, documented failure modes, and staff interpretation guidance.

Filtering Engine

Dimension	Detail
Objective	Apply hard gates to eliminate ineligible players
Inputs	Brief parameters + bio + finance + availability flags
Model type + explainability	Rule-based. Deterministic. Every exclusion states which gate failed.
Failure modes	Over-aggressive = missed targets. Under-aggressive = wasted time.
Staff interpretation	If excluded, the system says why. Override specific gates with logged justification.

Role-Fit Model

Dimension	Detail
Objective	Score statistical match to target role template
Inputs	League-adjusted per-90s + role template weights
Model type + explainability	Weighted cosine similarity. Full decomposition per metric.
Failure modes	Template may not reflect coach's actual needs if not updated. Garbage-in = garbage-out.
Staff interpretation	High role-fit = stats match what we want. Does NOT mean they will succeed. Video confirms.

League One Performance Model

Dimension	Detail
Objective	Current output quality vs. L1 peers at same role
Inputs	Per-90 metrics, form, consistency, opposition splits
Model type + explainability	Percentile ranking within role group. Show distribution.
Failure modes	Conflates dominant-team inflation with individual quality.
Staff interpretation	High score = producing now. Does NOT mean they produce at higher level.

Championship Projection Model

Dimension	Detail
Objective	Projected output, minutes share, and adaptation time at Championship
Inputs	Per-90 + league profiles + age + team strength + historical transitions
Model type + explainability	Hierarchical regression. ALWAYS outputs ranges (10th–50th–90th). Reports training sample size.

Failure modes	Survivorship bias. Small samples. Cannot capture intangibles. Sensitive to team-dominance adjustment.
Staff interpretation	Base case is best estimate. Range shows uncertainty. If range is very wide, the projection is unreliable — proceed with caution.

Availability Risk Model

Dimension	Detail
Objective	Probability of ≥75% match availability in next 12 months
Inputs	3yr availability, injury frequency/type/recurrence, minutes continuity, age
Model type + explainability	Logistic regression. Coefficients published and explainable.
Failure modes	Missing injury data for external players. Does not predict freak injuries. Does not capture destination S&C quality.
Staff interpretation	High Risk = data says historically unavailable. Medical MUST review before this influences shortlisting. Model is advisory to medical, never overrides.

Financial Value Model

Dimension	Detail
Objective	Fair-value fee band, wage-fit check, resale projection, replacement cost, VAR score
Inputs	Market value, age, contract, output, role, league, comparables, wage band
Model type + explainability	Quantile regression for value bands. Comparables listed. Assumptions stated.
Failure modes	TM values lag. Fees often undisclosed. Market dynamics unpredictable. Does not model agent friction or competitive bidding.
Staff interpretation	Model says what market typically pays for this profile. Actual fee depends on negotiation. Always present alongside: wage-ceiling fit, contract flexibility, replacement cost if player leaves, and deal timing.

Similarity & Alternatives Engine

Dimension	Detail
Objective	Find comparable or alternative targets
Inputs	Target player's league-adjusted per-90 profile as reference
Model type + explainability	Cosine distance on role-specific vectors, filtered by age/league/budget.
Failure modes	Statistical similarity does not mean stylistic similarity. Two similar profiles can play completely differently.
Staff interpretation	Use to generate alternatives, not to claim equivalence. Video review required for every alternative.

11.1 Note on Availability Model Scope

Honest framing: The availability risk model is useful but limited. It relies on public injury history and minutes data, which is incomplete for external players. It should be presented modestly: as an initial risk

screening that flags players for clinical review, not as a comprehensive medical assessment. Its strength is identifying chronic patterns (recurring muscle injuries, sustained unavailability). Its weakness is players with clean histories who then suffer new injuries. The model cannot predict those. Medical due diligence remains essential.

Section 12 — Outputs

12.1 MVP Outputs (Phase 1)

Four outputs only. These are what the recruitment team uses daily. Everything else is Phase 2.

Output	Who	When	Decision	Uncertainty Visible
Ranked longlist	Analyst + HoR	Per brief	Which players to investigate	Confidence tier. Projection ranges. Archetype classification.
Shortlist card (per player)	HoR + Scouts	During shortlisting	Deep profile per target: 6-layer breakdown, trajectory, risk, finance, pathway comparison	Per-layer confidence. Projection bands. Source flags. Internal comparison gap.
Target comparison sheet	HoR + Coach + Finance	Cross-functional review	Side-by-side 2–4 candidates on all layers	Where candidates differ most. Which differences are within confidence intervals.
Recommendation pack	SD / CEO / Board	Final approval	Complete evidence: stats, projection, risk, finance, scout reports, alternatives	All uncertainty disclosed. Ranges shown. Override history. Failure risk assessment.

12.2 Phase 2 Outputs

- Championship transition watchlist (monthly refresh)
- Emerging asset board (quarterly, with pathway comparison)
- Succession plan by role (pre-window planning)
- Post-window audit report
- Quarterly calibration report

Section 13 — Validation

Validation must answer football questions, not just model metrics. The core question: are we signing better players than we would without the model?

13.1 Outcome Framework

What counts as a hit, a miss, and a model-added win must be defined before deployment, not after.

Outcome	Definition	Evaluation Window	Role-Specific Criteria
Hit (performance)	Player's per-90 output in role-defining metrics is ≥50th percentile at destination level for ≥2 consecutive 6-month assessment periods	18 months	Role-defining metrics vary: goals+xG for strikers, tackles+interceptions for destroyers, progressive passes for playmakers
Hit (financial)	Player's market value at 18 months is ≥100% of acquisition cost (value retained or grown)	18 months	Strikers and wide attackers have more volatile values than CBs and GKs
Hit (availability)	Player available for ≥75% of matchdays in assessment period	12 months	Same threshold for all positions
Miss (performance)	Below 30th percentile at destination level for 2+ consecutive periods	18 months	As above, role-specific metrics
Miss (financial)	Market value declined >40% from acquisition cost AND performance does not justify retention	18 months	Applies equally across positions
Model-added win	Player was originally surfaced by the model (not by scout network or agent) AND qualifies as a performance hit	18 months	Tracks the model's unique contribution
Missed opportunity	Player was in a tracked league, matched an active brief, but was NOT surfaced by the model — and subsequently performed well elsewhere	Per window	Identifies systematic blind spots

13.2 Validation Methods

- **Temporal backtesting:** Train on all data before window W, predict outcomes for window W. Repeat for last 6 windows. Report precision@20, AUC, calibration.
- **Championship projection accuracy:** For players who moved L1 → Championship: did actual output fall within projected 10th–90th band? Target: ≥75% coverage.
- **Overperformance detection:** Did flagged Goals > xG overperformers regress in the following season? Target: ≥65% regression rate.
- **Age-value logic:** 22–24 signings retain or grow value in ≥60% of cases; 28+ signings in <20%.

- **Shortlist efficiency:** Higher conversion rate from longlist to signed player vs. pre-model process. Fewer wasted scouting trips.
- **False positive post-mortem:** Every signed player who underperforms gets a root-cause analysis: was it data gap, projection error, role mismatch, or accepted risk?
- **Adoption metrics:** Are scouts and decision-makers actually using the output? Track usage, satisfaction, and trust trajectory.

Section 14 — Governance

Area	Owner	Rule
Model logic	Head of Recruitment Analytics	Owns all model code and weights. Changes require written justification, version number, and HoR sign-off.
Role templates	First-Team Coach	Owns definitions. Reviewed quarterly. Every change logged with date, author, justification, and version.
Financial model	CFO + HoR Analytics	Co-owned. Board valuations require finance sign-off.
Medical override	Head of Medical	Absolute veto on medical grounds. Model availability risk is advisory only.
Filtering engine	HoR Analytics	Every gate rule is named and versioned. Gate changes require HoR approval.
Data sources	HoR Analytics	Source substitution requires reconciliation audit (<5% discrepancy) and documented approval.

14.1 Override Protocol

- **Every override is logged:** who, what, when, what the model recommended, what was decided instead, and why.
- **Override reason taxonomy:** (a) Tactical judgement (coach), (b) Financial constraint (finance), (c) Medical concern (medical), (d) Relationship/intelligence (HoR/scout), (e) Availability of alternatives, (f) Timing/window pressure. Every override must select a category.
- **Override outcomes tracked:** At post-window audit, overrides are compared against model's original recommendation. Were overrides more or less accurate? This builds the feedback loop.

14.2 Preventing Score Worship

- Composite scores NEVER presented without per-layer breakdown
- Training for all staff: the score starts a conversation, it does not end one
- A 72 is not better than a 68 if the difference is within the confidence interval
- Quarterly review: is the model being used as intended (decision support) or misused (decision replacement)?

14.3 Mandatory Change Logs

- Before each window: document any changes to weights, templates, gates, or sources since last window
- After each window: post-window audit pack (shortlists produced, progression rates, outcomes, overrides, data quality)
- Annually: full model performance review. Are models meeting success criteria? Presented to club leadership.

Section 15 — Implementation Roadmap

Phase 1: MVP (12–16 weeks)

One recruitment analyst can build a useful MVP. Decision-grade reliability depends on validation maturity, staff adoption, and disciplined governance.

Weeks 1–4: Data Foundation

- Full-field API-Football ingestion + FBref xG/xA + Transfermarkt values
- Per-90 normalisation, rolling aggregations, opposition quality tiers
- PostgreSQL schema, backfill 2–3 seasons

Weeks 5–8: Features + Gates

- League intensity normalisation, availability features, career trajectory
- Role classification, GBE calculation, confidence tiers
- Hard gate filtering engine, internal pathway comparison logic

Weeks 9–12: Model Training + Validation

- Role-Fit scoring, Championship Projection, Availability Risk, Financial Value models
- Temporal backtesting, calibration, bias audits
- Go/no-go: do models meet minimum thresholds?

Weeks 13–16: Deployment

- Four MVP outputs: longlist, shortlist card, comparison sheet, recommendation pack
- Override and prediction logging
- First live deployment with HoR review

Phase 2: Enrichment (Months 5–9)

- Wyscout deep scouting integration (zone-filtered, video-linked)
- Progressive actions, set piece attribution, pressing metrics
- Match-state analysis, versatility scoring, GK model
- Similarity engine, succession planning output
- First quarterly calibration report

Phase 3: Advanced (Month 10+)

- Physical tracking integration (when data access secured)
- Advanced financial modelling (replacement cost, wage optimisation)
- Manager-specific and competition-type segmentation
- Dashboard for non-technical staff

15.1 Staffing

- **Phase 1:** 1 analyst (Python/SQL + football recruitment understanding). This is a big lift. Analyst must be protected from ad-hoc requests during build.
- **Phase 2:** Same analyst + part-time Wyscout operator (existing scout with training).
- **Phase 3:** Ideally 2 people: one maintaining models/data, one on scouting integration and reporting.

15.2 Main Risks

- **Data quality:** Lower-league coverage can be patchy. Validate per league before trusting.
- **Staff turnover:** If the analyst leaves, system must be documented enough for a replacement. Version-controlled code, documented logic, data dictionary.
- **Adoption resistance:** Solve with transparency (show workings), humility (acknowledge limits), track record (publish validation).
- **Over-reliance:** If staff stop thinking critically, the model is being misused. Governance and override tracking prevent this.

Section 16 — Final Recommendation

16.1 Most Important Design Choices

12. **Dual-archetype assignment with blended weighting.** Reflects the reality that signings serve multiple purposes. Prevents over-simplified evaluation.
13. **Championship projection as a first-class three-dimensional output.** Performance, minutes share, and adaptation time. This is what separates Stockport's model from a survival-focused approach.
14. **Three-class gate system.** Absolute exclusions, strong cautions, and sign-off exceptions. Prevents both over-rigid filtering and gate-free permissiveness.
15. **Tactical fit as a score cap.** A player who does not fit the system cannot rank highly regardless of statistical output. This is the most football-aware design choice in the model.
16. **Process control with defined stage transitions.** Required inputs before briefs go live. Minimum scouting before cross-functional review. Explicit conditions for longlist-to-shortlist. Defined failure criteria.
17. **Entity-level source hierarchy.** Different data entities have different sources of truth. No blanket canonical philosophy.

16.2 What to Cut

- Competitive intelligence (predicting rival bidders). Unvalidatable.
- Second-order squad impact. No ground truth.
- Automated sell-on timing. Too uncertain beyond 2 years.
- Physical tracking until data access is secured.
- Any feature not traceable to one of the five decisions in Section 1.

16.3 What to Protect

- **Validation framework.** Without it, the club cannot know if the model helps.
- **Human override protocol.** Without it, the model becomes unchallenged authority.
- **Pathway integration.** Without it, asset-building is disconnected from recruitment.
- **Process control.** Without defined stage transitions and failure criteria, the framework is informative but procedurally soft.
- **Transparency.** Every score decomposable. Every projection a range. Every risk flag challengeable. The moment it becomes a black box, trust dies.

Final Word

This model does not make Stockport County smarter. It makes the people at Stockport County better informed. The intelligence is in the decisions those people make with better evidence, clearer uncertainty, and sharper questions. The model is the infrastructure. The people are the advantage.